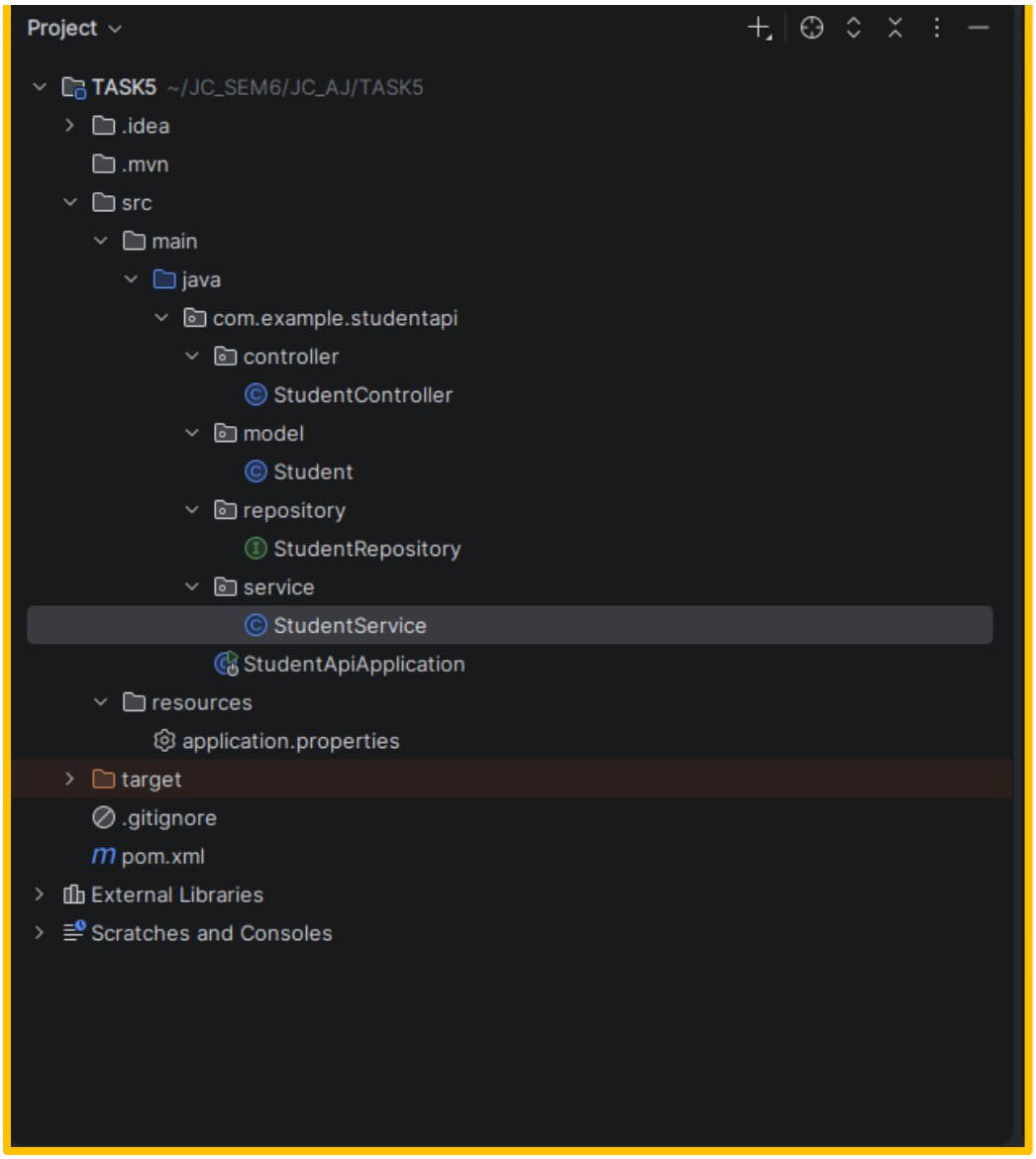


Above is project file structure

```java
package com.example.studentapi.controller;

import com.example.studentapi.model.Student;
import com.example.studentapi.service.StudentService;
import org.springframework.web.bind.annotation.*;

import java.util.List;

@RestController
@RequestMapping("/students")
public class StudentController {

    private final StudentService service;  4 usages

    public StudentController(StudentService service) {
        this.service = service;
    }

    // GET all students
    @GetMapping
    public List<Student> getAllStudents() {
        return service.getAllStudents();
    }

    // POST add student
    @PostMapping
    public Student addStudent(@RequestBody Student student) {
        return service.addStudent(student);
    }

    // GET search by name
    @GetMapping("/search")
    public List<Student> searchStudents(@RequestParam String name) {
        return service.searchByName(name);
    }
}
```

Above is StudentController.java

```java
package com.example.studentapi.model;

import jakarta.persistence.*;

@Entity
public class Student {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;
    private String email;

    public Student() {}

    public Student(String name, String email) {
        this.name = name;
        this.email = email;
    }

    public Long getId() { return id; }

    public void setId(Long id) { this.id = id; }

    public String getName() { return name; }

    public void setName(String name) { this.name = name; }

    public String getEmail() { return email; }

    public void setEmail(String email) { this.email = email; }
}
```

Above is model Student.java

Java file is located outside of the module source root, so it won't be compiled

```java
package com.example.studentapi.repository;

import com.example.studentapi.model.Student;
import org.springframework.data.jpa.repository.JpaRepository;
import java.util.List;

public interface StudentRepository extends JpaRepository<Student, Long> {  3 usages

    List<Student> findByNameContainingIgnoreCase(String name);  1 usage
}
```

Above is StudentRepository.java

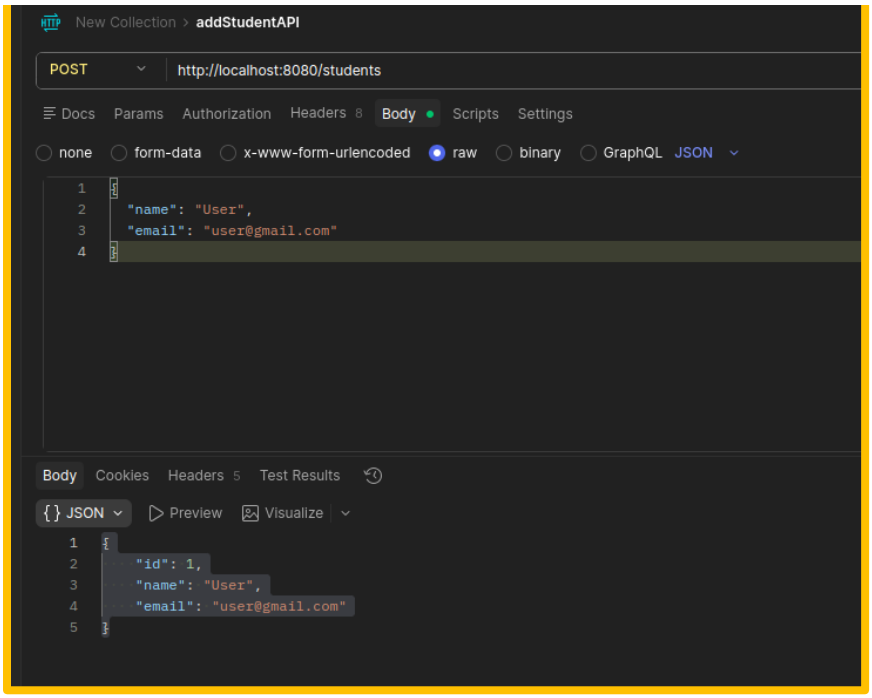
```java
package com.example.studentapi.service;

import com.example.studentapi.model.Student;
import com.example.studentapi.repository.StudentRepository;
import org.springframework.stereotype.Service;

import java.util.List;

@Service  3 usages
public class StudentService {

    private final StudentRepository repository;  4 usages

    public StudentService(StudentRepository repository) {
        this.repository = repository;
    }

    public List<Student> getAllStudents() {  1 usage
        return repository.findAll();
    }

    public Student addStudent(Student student) {  1 usage
        return repository.save(student);
    }

    // 🔥 THIS METHOD WAS MISSING
    public List<Student> searchByName(String name) {  1 usage
        return repository.findByNameContainingIgnoreCase(name);
    }
}
```

Above is StudentApiApplication.java

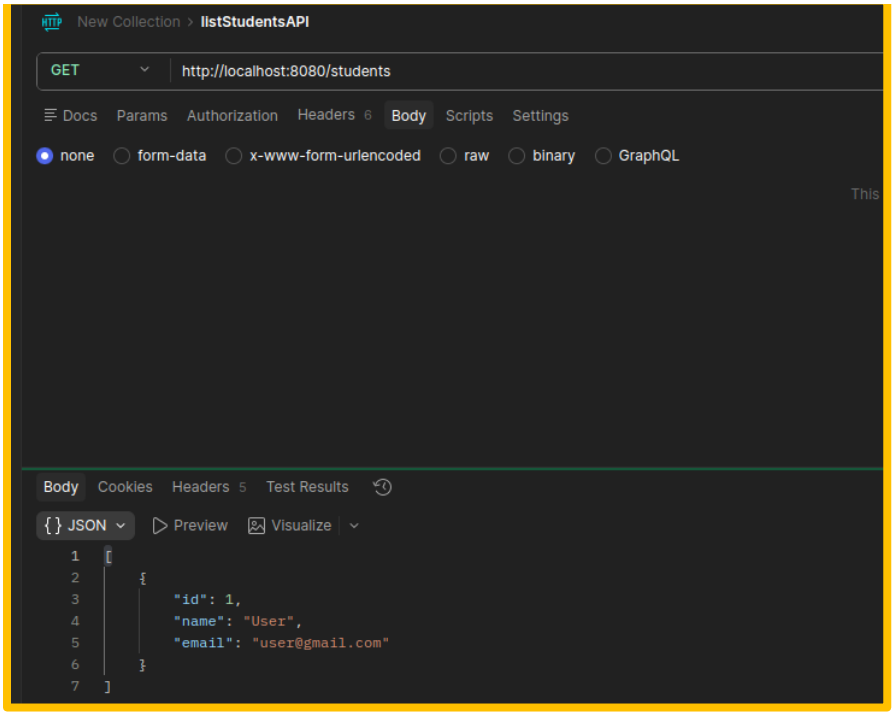
```xml
<parent>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-parent</artifactId>
  <version>3.2.0</version>
</parent>
<dependencies> Add Starters...
  <dependency>
    <groupId>junit</groupId>
    <artifactId>junit</artifactId>
    <version>3.8.1</version>
    <scope>test</scope>
  </dependency>
  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-web</artifactId>
  </dependency>

  <!-- JPA -->
  <dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-data-jpa</artifactId>
  </dependency>

  <!-- H2 DB -->
  <dependency>
    <groupId>com.h2database</groupId>
    <artifactId>h2</artifactId>
    <scope>runtime</scope>
  </dependency>
</dependencies>
<build>
  <plugins>
    <plugin>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-maven-plugin</artifactId>
    </plugin>
  </plugins>
</build>
```

Above is pom.xml code.

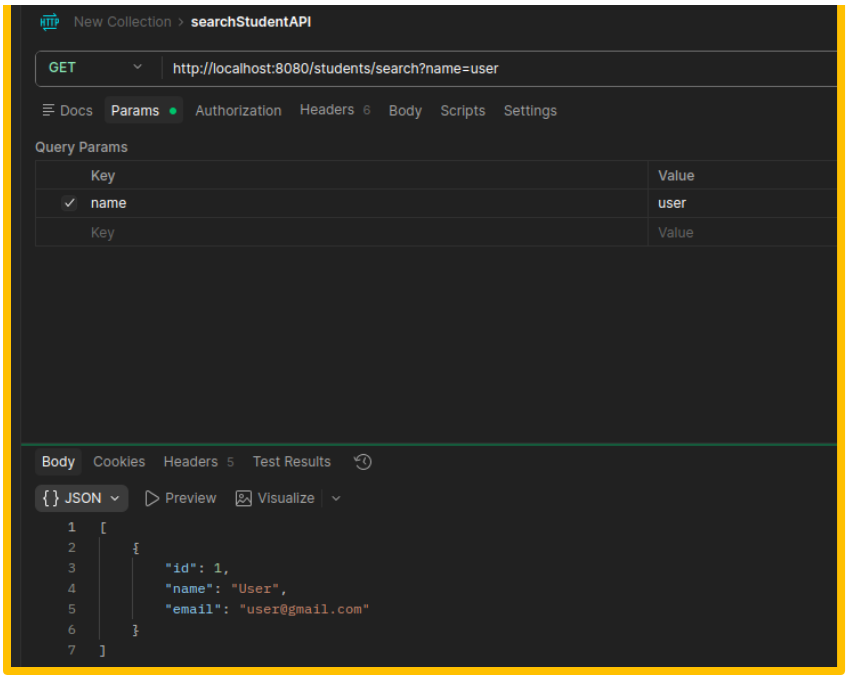
Output :



Above is addStudentAPI



Above is listStudentsAPI



Above is searchStudentAPI